

HILL CITY, KS, USA

WMO: 724655

Lat: 39.376N

Lon: 99.830W

Elev: 2188

StdP: 13.57

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 93990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 7.2	(d) -5.8	(e) 4.4	(f) 4.7	(g) -1.0	(h) 5.6	(i) 10.2	(j) 30.3	(k) 36.0	(l) 26.9	(m) 38.2	(n) 7.7	(o) 0	(p) 0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.9	(c) 101.5	(d) 71.0	(e) 98.6	(f) 71.0	(g) 95.1	(h) 70.8	(i) 75.7	(j) 90.8	(k) 74.4	(l) 89.7	(m) 73.1	(n) 88.6	(o) 13.3	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	128.3	82.2	70.3	120.9	80.5	68.9	115.3	79.4	40.7	90.2	39.3	89.8	38.1	88.6	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.8	24.6	21.1	-5.7	106.4	4.4	3.0	-8.9	108.6	-11.5	110.3	-14.0	112.0	-17.2	114.1		
			-6.5	77.9	4.1	1.5	-9.4	79.0	-11.8	79.8	-14.1	80.7	-17.1	81.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	30.7	33.2	43.3	52.5	63.1	74.4	80.2	76.9	69.0	54.9	41.9	31.3
	DBStd	19.86	10.97	11.56	10.95	9.65	8.68	7.49	6.37	5.91	8.68	9.85	9.81	10.38
	HDD50	2347	598	480	261	82	7	0	0	0	1	61	276	581
	HDD65	5300	1062	892	674	387	141	13	1	1	57	335	693	1044
	CDD50	3953	1	8	53	157	413	731	936	835	571	213	33	2
	CDD65	1429	0	0	1	12	82	293	472	371	176	22	0	0
	CDH74	17733	1	10	69	285	1147	3631	5900	4140	2131	394	25	0
	CDH80	9188	0	1	10	84	465	1897	3399	2190	1014	126	2	0
	WSAvg	9.7	8.9	9.8	10.7	11.8	10.4	10.0	9.0	8.3	9.3	9.6	9.2	9.0
	PrecAvg	21.9	0.5	0.6	1.3	2.0	3.8	3.0	3.5	3.0	2.2	1.5	0.7	0.6
Precipitation	PrecMax	32.5	1.4	1.6	3.6	5.8	12.6	7.1	12.7	7.0	6.6	6.8	2.6	4.6
	PrecMin	12.0	0.0	0.0	0.0	0.1	1.0	0.9	0.5	0.4	0.7	0.0	0.0	0.0
	PrecStd	4.6	0.4	0.5	1.0	1.2	2.4	1.5	2.4	1.8	1.3	1.5	0.6	0.9
	PrecStd	4.6	0.4	0.5	1.0	1.2	2.4	1.5	2.4	1.8	1.3	1.5	0.6	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	73.5	81.0	88.1	95.9	104.8	106.1	102.4	99.8	90.4	77.7	64.4
		MCWB	47.9	50.1	56.4	60.0	65.0	69.7	71.5	70.5	68.6	64.9	54.3	47.8
	2%	DB	60.6	64.7	74.9	81.7	90.0	98.9	102.4	99.1	94.7	82.8	71.5	58.4
		MCWB	44.2	47.4	54.0	59.3	65.8	70.4	71.5	71.1	68.7	61.7	52.8	44.6
	5%	DB	54.0	57.7	69.6	76.3	85.1	94.9	99.4	95.4	90.5	78.4	64.8	53.4
		MCWB	41.5	44.3	52.4	57.9	64.9	70.3	72.1	71.2	68.0	59.9	49.9	42.3
	10%	DB	47.9	52.1	63.4	71.0	80.8	90.7	96.0	91.5	86.0	72.7	59.3	47.7
		MCWB	38.4	41.6	50.1	55.5	63.5	69.6	72.0	70.9	66.8	57.9	46.7	39.0
	0.4%	WB	49.1	52.4	62.1	64.4	71.7	76.5	77.7	77.2	74.4	69.0	58.3	50.7
		MCDB	64.8	65.0	71.9	77.4	84.6	92.6	93.5	91.4	87.9	81.9	68.9	61.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.1	48.8	58.0	61.9	69.3	74.2	76.1	75.2	72.0	65.9	54.6	45.5
		MCDB	58.7	63.8	68.8	75.3	82.8	90.0	91.8	89.3	86.5	76.3	66.1	57.1
	5%	WB	42.0	45.3	54.2	59.4	67.2	72.5	74.8	73.6	70.1	62.6	51.5	42.3
		MCDB	53.1	57.0	65.1	73.1	80.3	88.5	90.6	88.2	84.3	73.7	63.6	52.4
	10%	WB	38.8	41.8	50.4	56.7	65.3	71.0	73.5	72.3	68.4	59.4	48.0	39.2
		MCDB	47.3	51.6	63.8	69.4	77.4	86.5	89.7	86.6	81.7	70.0	59.3	47.5
Mean Daily Temperature Range	5% DB	MDBR	24.5	24.8	27.6	26.6	25.8	26.3	25.9	25.0	26.6	27.0	26.4	23.7
		MCDBR	35.2	36.7	37.6	35.5	32.7	31.7	30.7	30.2	31.2	35.0	36.0	32.6
	5% WB	MCWBR	21.8	21.6	20.5	17.4	13.9	10.4	9.0	9.8	11.7	16.6	19.9	20.7
		MCDBR	33.6	34.5	32.4	30.3	27.4	27.8	26.7	26.1	26.8	28.5	33.0	31.1
	5% WB	MCWBR	21.3	21.0	19.3	16.5	13.7	11.2	9.7	10.1	11.5	15.1	19.0	20.4
		MCDBR	33.6	34.5	32.4	30.3	27.4	27.8	26.7	26.1	26.8	28.5	33.0	31.1
Clear-Sky Solar Irradiance	taub		0.270	0.276	0.306	0.339	0.364	0.380	0.404	0.386	0.350	0.306	0.276	0.264
	taud		2.521	2.518	2.449	2.384	2.394	2.382	2.354	2.393	2.433	2.545	2.548	2.534
	Ebn at Noon		292	305	303	297	290	284	276	279	284	289	287	285
	Edh at Noon		23	27	32	37	37	38	39	36	33	26	23	22
All-Sky Solar Radiation	RadAvg		777	1036	1425	1755	2019	2268	2228	1942	1600	1166	861	677
	RadStd		58	94	116	93	129	121	91	117	91	119	61	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page